Alvin Christian Nataputra

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Research interests

Reliable machine learning systems: validity and leakage control in time-series ML, distribution shift, and how autonomous agents stay grounded over long horizons. Applied work on Indonesian-language text and Indonesian equity markets, where data is thin and regimes change.

Education

IPB University (Bogor Agricultural University) Bogor, Indonesia

Bachelor of Statistics and Data Science • *August 2019 – August 2023* • *GPA 3.83/4.00*

Thesis: suicide-ideation detection in Indonesian user-generated text (FastText + LSTM, later an IndoBERT comparison). A 2026 leakage audit of the same split is reported under Research; the work is research/educational and is not a clinical tool.

Research

Reproducible evaluation of an IDX equity signal pipeline 2025–2026

Independent research; ensemble ML for the Indonesia Stock Exchange

- Audited an existing evaluation of a LightGBM/XGBoost signal pipeline and found four defects that invalidated its reported performance: the final model was trained on rows spanning the test window, walk-forward early stopping used the fold's own test set, the feature scaler was fitted on the whole series before splitting, and the embargo was five calendar days against a ten-bar label horizon with no purging of overlapping labels.
- Rebuilt the evaluation with purged and embargoed folds, fold-local scaling, average-uniqueness sample weights for overlapping labels, and buy-and-hold and momentum baselines under identical IDX transaction costs.
- Reported result is negative and is the point of the work: the model returns +0.12% / +7.62% / +4.73% across three configurations, against +14.17% for buy-and-hold and +20.71% for the index; walk-forward accuracy of 0.34–0.36 sits below the 0.53 majority-class baseline on 0 of 16 tickers (sign-test $p = 3 \times 10^{-5}$). Median time in market is 3% of test days.
- Committed a full run manifest — per-file data SHA-256, exact split dates, seeds, configuration, environment lockfile, trade ledger and predictions — so the result is re-derivable rather than asserted. github.com/apricitea/project-aurum/tree/main/research

Duplicate leakage in a scraped Indonesian mental-health corpus 2026

Independent research note; evaluation of the undergraduate thesis corpus

- Measured that a random 80/20 split of a 10,128-post scraped X corpus placed 739 of 2,026 validation posts (36.5%) in both halves, because 2,161 posts are exact duplicates and the split was drawn before they were removed.
- Re-scored the committed IndoBERT predictions on the 1,287 unseen rows: positive-class F1 0.877 [0.845, 0.904], down from 0.910. Compared against four classical baselines trained on the same data; the strongest (character n-gram logistic regression) reaches 0.787, so the remaining transformer advantage is +0.090 [+0.049, +0.131] (McNemar $p = 6.3 \times 10^{-4}$).
- Calibration ECE 0.046; 90% recall requires reviewing 21.8% of the corpus at 83.6% precision. Paper, tests, CI, committed run manifest. github.com/apricitea/text-suicide-ideation-detection

Bahasa Indonesia corpus: collection pipeline and rights-cleared release 2026

Public dataset construction with a provenance gate

- Built and operate a weekly pipeline collecting Indonesian text (Wikipedia extracts and news sources), with an explicit release rule: a record is published only if its source maps to a licence that can be asserted *and* it carries source URL, revision ID, title and retrieval timestamp.
- Released 552 attributable records under CC BY-SA 4.0 with per-record attribution; excluded roughly 2,200 news-derived and 30,000 unattributed records rather than publish them without a licence. huggingface.co/datasets/apricitea/bahasa-indonesia-corpus

Market-regime dating for IDX equities 2024

Implemented a censored local-extrema algorithm in the style of Pagan & Sossounov: candidate turning points are filtered by strict peak/trough alternation and phases must clear minimum duration and amplitude

thresholds, so no arbitrary fixed lookback window is assumed. Validated by a test suite and shown to isolate the 2020 COVID drawdown as a distinct bear phase on BBRI without supervision.

Causal effect estimation in telco churn 2023

Extended an XGBoost churn model (SHAP-explainable) with an inverse-probability-of-treatment-weighting estimator and bootstrapped confidence intervals, to separate correlation from causation for product-adoption decisions. Call Center adoption causally reduced churn and increased CLTV; video and MyApp adoption increased monthly purchase but their CLTV effects were indistinguishable from zero — which changes the recommendation from "push adoption" to "fix the product first".

Telecommunications CVM analytics and ML lab 2026

Idempotent DuckDB warehouse with a data-quality gate that aborts before touching downstream tables, two independently calibrated risk models (churn AUC 0.823; late-payment AUC 0.628), and a test that enforces the forward-looking construction of the late-payment label.

Experience

XLSmart — Data Scientist, Customer Value & Experience Aug 2024 – present

Jakarta, Indonesia • *formerly XL Axiata* • *third-largest Indonesian telco, 80M+ subscribers*

- Data science and analytics partner for XL Prioritas (postpaid), Customer Experience and Go-to-Market teams; own end-to-end analytics from opportunity identification through segmentation, targeting and impact measurement.
- Built churn-driver and high-risk-segment analysis that shaped retention initiatives; the resulting programme was reported to improve customer retention by 25%.
- Led end-to-end analytics for targeted campaigns, enabling commercial teams to optimise targeting and measure incremental revenue.
- Built an agentic-AI analytics layer used by senior leadership to monitor key metrics, identify performance drivers and receive recommendations.
  Stack: Python, SQL, BigQuery, Databricks, Snowflake, Tencent DLC/WeData.

Asta Protek Jiarsi — Data Scientist, IT consulting Sep 2023 – Jun 2024

Jakarta, Indonesia • *consulting and compliance services*

- Delivered ML models, segmentations and dashboards for anti-money-laundering, fraud detection and enterprise analytics, working with client teams to production.
- Standardised ETL workflows and Docker-based ML deployment pipelines, removing a recurring class of production incidents and improving reproducibility.
- Applied SHAP to explain model predictions, supporting business interpretation and regulatory compliance for client ML solutions.
  Stack: Python, Poetry/uv, PostgreSQL, SQL Server, Docker.

Indosat Ooredoo Hutchison — Data Science & Analytics Intern May – Aug 2023

Jakarta, Indonesia • *second-largest Indonesian telco, 100M+ subscribers*

Automated recurring business analyses from spreadsheets into dashboards and scheduled reports, cutting manual reporting workload by 20%; ran deep-dive statistical testing on product and campaign performance, identifying Pareto contributors.

Bank Rakyat Indonesia (BRI) — Analytics Intern Oct 2022 – Apr 2023

Jakarta, Indonesia • *state-owned bank*

Built a non-performing-loan risk model reaching 60%+ recall at 70% precision; produced geospatial analyses supporting regional coverage optimisation (10%+ operational efficiency improvement).

Technical

Languages Python, SQL, R, shell

Machine learning gradient-boosted trees (LightGBM, XGBoost), calibrated classification, clustering, feature engineering, model evaluation, explainability (SHAP), transformers (BERT/IndoBERT), LSTM, FastText

Statistics & experimentation A/B testing, control-group design, hypothesis testing, causal inference (IPTW, propensity scores), time-series validation (walk-forward CV, purging, embargo)

Engineering & reproducibility Docker, uv, pytest, GitHub Actions CI, Git, PostgreSQL, DuckDB, Qdrant

Data platforms BigQuery, Databricks, Snowflake, Tencent WeData/DLC

Natural languages Indonesian (native), English (Cambridge C1 Advanced in preparation; examination targeted 2027)